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Secular Variation in the Pacific Ocean Region

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Summary

Recent geomagnetic data shows a region centred on the Pacific Ocean in which secular variation of the geomagnetic field is lower than elsewhere. Knowledge of the geographical and/or temporal permanence of this feature can aid in understanding the deep structure and history of the Earth and has obvious implications with respect to the common assumption of a spherically symmetric earth. A literature survey has been carried out to find palaeomagnetic data which meet sufficient criteria to be considered measures of palaeosecular variation (PSV) of the geomagnetic field. Analysis of these data indicates that the Pacific PSV low has existed throughout the Cenozoic Era and that perhaps it was slightly more pronounced during Tertiary than during Quaternary times. These results are dependent on the size of the area included in the 'Pacific Region'; excluding data from the margins of the Pacific Ocean increases the PSV contrast but the small numbers of remaining Pacific Region data make the results statistically questionable.

1. Introduction

In seeking a greater understanding of the Earth's interior through study of the secular variation of the geomagnetic field, it is important to learn more about the distribution of the field over the surface of the globe throughout geologic time.

The present study was undertaken after the collection of palaeomagnetic data for palaeosecular variation (PSV) analysis in Alaska (Bingham 1971; Bingham & Stone 1972) made it clear that determining the extent and permanency of the region of low secular variation centred on the Pacific Ocean (Doell & Cox 1971; and others) was a complex problem, due primarily to the extreme variability of available data.

The palaeomagnetic method of PSV measurement involves obtaining a series of spot readings of the palaeomagnetic field at a particular location, spanning a sufficient length of time to include the PSV spectrum. Based on modern-day geomagnetic field measurements, as well as on archeomagnetic and palaeomagnetic measurements, secular variation of the field is generally considered to have a spectrum ranging from about 10 to 10^4 years (Cox & Doell 1964). The most usual formations used for palaeomagnetic measurement of PSV are sequences of lava flows, as an individual lava may be considered to represent an instant on the geologic time scale. It is generally considered that at least 20 flows should be sampled for statistically reliable results, but for practical reasons 10 flows is often used as a lower limit.

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Due to the inadequacies of present dating techniques, it is necessary to treat the palaeomagnetic results statistically, taking the dispersion of a set of palaeofield directions (PFD's) or their associated virtual geomagnetic poles (VGP's) (Cox & Doell 1960) to be a measure of PSV. The dispersion parameters commonly used are

$$k = \frac{N-1}{N-R}, \tag{1}$$

the Fisher (1953) precision parameter, where R is the length of the resultant vector obtained by adding N unit vectors, and the angular standard deviation, given by

$$\delta = \cos^{-1}\frac{R}{N} \tag{2}$$

(Wilson 1959), or by

$$\Delta = \sqrt{\frac{2}{k}} \text{ radians} = \frac{81}{\sqrt{k}} \text{ degrees} \tag{3}$$

(Creer 1962a, b; and others). Δ and δ are not necessarily equivalent, but numerically they usually differ only slightly (cf. Bingham 1971). k, Δ, and δ may be applied equally well to a group of PFD's or to their corresponding VGP's.

The question as to whether or not PSV analysis using PFD statistics is preferable to using VGP statistics is a perplexing one. Since the VGP is only a mathematical construct with no real physical meaning, due to the fact that VGP's corresponding to PFD's do not necessarily define true magnetic poles, it might seem that using PFD's would be the more fundamental method. On the other hand, the VGP analytical method allows an easier separation of the effects of dipole wobble and non-dipole sources, since the angular dispersion of VGP's due to dipole wobble is invariant with respect to latitude (Cox 1962; Cox & Doell 1964). Cox (1970) in a careful analysis of the VGP–PFD problem concludes that the VGP method is preferable, but that there is not enough world-wide data to use his technique for complete separation of non-dipole from dipole components.

Because the palaeosecular variation measurement process involves statistical parameters which assume a Fisher, or at least a circular distribution, a further problem arises, inasmuch as a circular distribution of PFD's transforms into an oval distribution of VGP's while a circular distribution of VGP's transforms into an oval distribution of PFD's. The various sources of scatter will produce either circular or oval distributions depending on how they are calculated. Experimental errors will produce a circular scatter of PFD's. Secular variation itself will consist of both circular and oval components, as those parts of PSV due to non-dipole sources may be expected to produce a circular scatter of PFD's with the dipole sources giving a circular VGP distribution. Thus it is not immediately clear whether Fisher (1953) statistics, which require an axially symmetric distribution, are appropriate in either case.

In addition to the above problems, there is also a magnitude effect, associated with the PFD–VGP transformation. Cox & Doell (1960, 1964) show that the ratio of the angular standard deviations varies from

$$\frac{\delta_{pfd}}{\delta_{vgp}} = \frac{1}{2}$$

at the poles to

$$\frac{\delta_{pfd}}{\delta_{vgp}} = \frac{3}{2}$$

at the equator. These relations are in part responsible for the apparently opposite latitude dependence of dispersion of PFD's and VGP's derived from the present field discussed by Creer (1962a) and Cox & Doell (1964), respectively.

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For the study of secular variation of the geomagnetic field it would clearly be advantageous to separate the circular and oval components of scatter from each other. Unfortunately, no statistical formulation exists for conveniently resolving these components from presently available data.

Since it is probable that random errors and non-dipole components generally predominate in the available PSV measurements, and to conform to the majority of other published results, we have chosen to do comparative analysis almost exclusively on PFD's instead of VGP's.

Since, generally, several samples are obtained from a single lava flow to obtain a flow mean PFD, a distinction must be made between the within-flow precision, k_ω, the between-flow precision, k_β, and the measured between-flow precisions, k_T, biased by both k_ω and k_β. These factors, or their equivalents Δ_ω, Δ_β, Δ_T, are usually separated using analysis of variance technique (Watson & Irving 1957).

2. The Pacific secular variation low

If one looks at recent secular variation of the geomagnetic field on a global scale by examining almost any type of geomagnetic analysis maps, one feature is immediately evident: secular variation of the present-day geomagnetic field is less in the hemisphere containing the Pacific Ocean than it is in the remaining part of the world (Fisk 1931; Fleming 1939; Chapman & Bartels 1940; Vestine *et al.* 1947; Runcorn 1956; Depietri 1961; Vestine & Kahle 1966; Doell & Cox 1971; and others). Although the boundary between areas of high and low secular variation is difficult to define, there is no doubt of its existence.

The question is immediately raised as to whether or not the Pacific low is a permanent feature of the secular variation field in time as well as geographically. Cox & Doell (1964) and Doell & Cox (1961, 1965) conclude, after an analysis of Pacific palaeomagnetic data from the Galapagos Islands to Alaska, that the non-dipole component of PSV has been subdued in the Pacific during the last 10^6 years. However, Tarling's (1967) studies in Hawaii, Samoa, and the Cook Islands indicate that PSV has not been anomalously low during the last 5 My when time averages of 10^5 and 10^6 y are considered.

One can devise various models to explain the low PSV values in the Pacific which may treat the problem as either a transient or a permanent phenomenon. It is clear from analyses of the present field that this difference in distribution of PSV cannot be caused by the broad differences in crustal structure which exist between the 'oceanic' Pacific hemisphere and the essentially 'continental' non-Pacific hemisphere, but rather must have its origin in deep seated variations in the structure, or in physical processes occurring within the mantle and/or the core. If the PSV differences do in fact persist over geologic time, this may require a basic reformulation of the physics of the Earth's interior which formerly has relied on a spherically symmetric earth (Runcorn 1956).

Cox (1962) delineates three possible models to explain the Pacific PSV low, which consider only the non-dipole part of PSV, since that part due to movement of the main dipole itself (wobble, etc.) will be detected equally world-wide.

(1) Non-dipole sources of PSV do not form under the Pacific region.

(2) Non-dipole sources of PSV are present under the Pacific but their effects are screened from observation at the Earth's surface by highly conducting, possibly fluid, layers in the mantle.

(3) Non-dipole sources may be present or absent beneath the Pacific but at the present time there are none in this region.

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Given the present uncertainties surrounding palaeomagnetic methods of PSV measurement and the conflicting results of some researchers, there is a possibility that the present PSV distribution is due to a chance arrangement of non-dipole sources within the Earth, so the last of these models (3), cannot be ruled out. Until more definitive palaeomagnetic results become available, however, this explanation appears to be begging the question and it is fruitful to look for models which provide for a long-term low PSV in the Pacific region.

A number of workers have attempted to explain model (2). Creer (1963) hypothesized magnetic shielding of the PSV field by an asymmetric mantle. He extended Elsasser's (1963) cold accretion Earth origin hypothesis to show how, during early stages of geoformation, drops of molten iron sinking through the mantle toward the core would sweep up iron particles along the way. This scheme could leave parts of the mantle 'unswept' of iron providing the necessary shielding of PSV.

A central problem in the magnetic shielding hypothesis is to account for the amount of shielding which is apparently necessary. Unreasonably high conductivity and/or great thickness of the shielding layer are required. McDonald (1957), Hide & Roberts (1961), and Currie (1967) find from theoretical calculations that, taking a world average, the mantle has a time constant of about four years and that it acts as a low-pass filter which effectively blocks out magnetic variations with shorter time scales. Ball, Kahle & Vestine (1968) find a time lag of 5–8 y for transmission of magnetic disturbances from the core to the surface. Cox & Doell (1964) point out that conductivities large enough to attenuate secular variation in the 10^2 to 10^3 y range would reduce short period (10 y) variations below noise level; but short period variations have indeed been observed in Hawaii.

Any well-defined shielding layer thick enough to produce the requisite PSV attenuation beneath the Pacific should be thick enough to be detectable seismically. In a study of attenuation of different frequency components of *P* waves diffracted along the core–mantle boundary, Alexander & Phinney (1966; Phinney & Alexander 1966) find systematic differences between the central Pacific and the other areas they studied. On the basis of lengthy calculations from their data, they postulate that in the lower mantle beneath the Pacific there is a region between 30 and 160 km thick with a reduced shear velocity and an increased density. What this means in terms of shielding is a matter for speculation, but it could be indicative of residual iron such as that hypothesized by Creer (1963).

Cox's (1962) model (1) advanced to explain the Pacific SV low, namely that non-dipole sources do not form under the Pacific, is perhaps easier to justify physically than the shielding hypothesis, though it is equally difficult to prove. By this hypothesis there are lateral variations in the sources that produce PSV rather than variations in the transmission parameters of the medium through which disturbances must travel. Cox & Doell (1964) propose that temperature differences in the lower mantle may be sufficient to affect the convection currents in the core which produce the non-dipole field. Citing MacDonald's (1963) calculations showing 50° to 150 °C higher temperatures 1000 km beneath the oceans than beneath continents, they propose that under the essentially 'oceanic' Pacific hemisphere the lower mantle may be hot enough to suppress near-surface core convection by causing reduced lateral as well as radial temperature gradients. Thus non-dipole features of the field would decay upon moving into the Pacific area.

Vestine & Kahle (1966) propose the existence of non-uniform core surface velocities with a general downflow of core material beneath the Pacific Ocean. By carrying magnetic disturbances caused by local eddies deeper into the core, lower values of PSV would be produced than in areas where core motions were predominantly horizontal or upwards, and where there is less shielding core material above them.

Doell & Cox (1971), following Hide (1966, 1967) also suggest a different way to

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cause core eddy currents sufficiently asymmetric to produce the Pacific secular variation low. In comparing motions of the core with those of the atmosphere, they note that while both are essentially axially symmetric when averaged over sufficient time spans, both have more or less permanent perturbations of this symmetry due to the topographic relief and temperature variations on the solid–fluid interface. As with the production of standing waves in the atmosphere due to mountains, it is conceivable that there may be a broad topographic high or low in the core–mantle interface associated with the Pacific hemisphere which is sufficient to produce an anomalous lack of eddy currents in this region.

3. Palaeosecular variation data

In an attempt to learn more about the world-wide distribution of palaeosecular variation, a survey was made of palaeomagnetic literature to select existing data which can be considered to be a measure of PSV (Tables 1 and 2). The compilations of palaeomagnetic data of Irving (1964) and McElhinny (1968a, b, 1969, 1970) have been used as a starting point for the survey. Where available, the original source literature has been checked in order to estimate the relative reliability of the data. An attempt has been made to review every paper through May 1970 since the last of these compilations but no claim at completeness is made.

The following criteria have been used in selecting data.

(1) Only data from rocks of the Cenozoic Era have been considered.

(2) A minimum of $N = 10$ flows or sites sampled within a volcanic sequence was required. Some exceptions were made in order to include data from isolated regions of the Earth.

(3) Studies with α_{95} confidence limits greater than 15° about the mean vector direction were rejected on the basis that either the errors from various sources were large, or that the sequence included values resulting from unusually large field excursions. This criterion tended to eliminate many data sets with small N but still large enough to be allowable by (2) above.

(4) If the mean VGP of a set of data differed from the present geographic pole by more than 40° the data were rejected on the grounds that they probably represent a large-scale field excursion. Since such excursions seem to be rare, their inclusion in palaeosecular variation studies could lead to abnormally high values for PSV (Doell 1970b).

(5) Only data which appeared by suitable tests to represent stable palaeomagnetic results were selected. In all the more recent papers, tests such as alternating field, thermal demagnetization, or storage were carried out. In a few of the earlier papers, stability was indicated by such tests as divergence of the mean from the present field, the presence of reversed and normal magnetization directions approximately 180° apart, and the tightness of within-flow distribution (Irving 1964). In fact the inclusion of data from these latter papers does not significantly affect the results.

(6) For inclusion in the compilation, enough data had to be given to allow the usual statistical parameters to be calculated.

The results have been roughly divided into two geographic groups; data from the Pacific low region (Table 1) and data from outside this region (Table 2); within each region the data are listed more or less according to present day geographic latitude. The Pacific region includes Alaska, Japan, the west coast of North America, eastern Australia, New Zealand and all the Pacific Islands. This division is admittedly

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artificial as it is difficult to sharply delineate areas of 'high' and 'low' secular variation. As described below, the region of the Pacific Ocean in which PSV is low is perhaps somewhat smaller than the geographic division used for Table 1.

Making an additional subdivision of the data in Tables 1 and 2 according to geologic age represented, Tertiary and Quaternary, Δ_T has been plotted against present-day latitude of the sampling sites (Figs 1–4). Palaeolatitudes have not been used in these diagrams; for only a very few of the points is the palaeolatitude significantly different from the present-day latitude and the overall picture would be changed little by using them. Like the division into geographic regions described above, the separation of data from such widely varying sources into 'Tertiary' and 'Quaternary' groupings is also somewhat arbitrary. Many of the dates are but

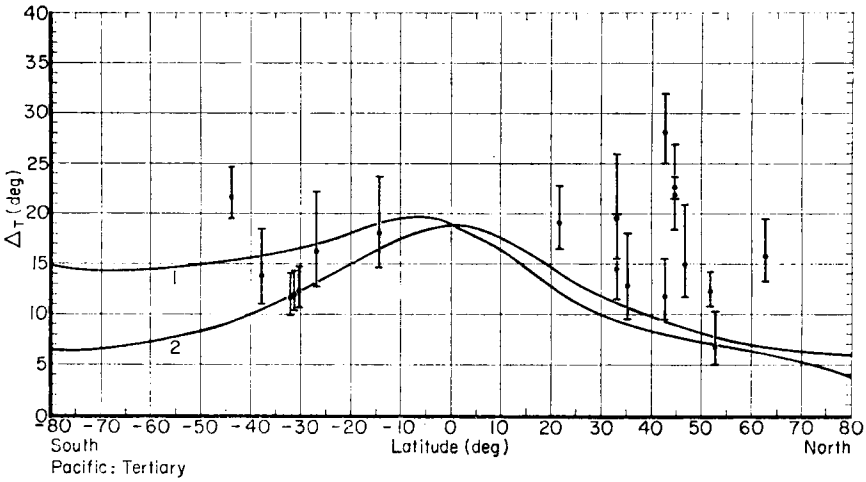


FIG. 1. Tertiary palaeosecular variation data for the Pacific region. Error bars calculated according to Cox (1969a). Data from Table 1, Curves 1 and 2 from Creer (1962).

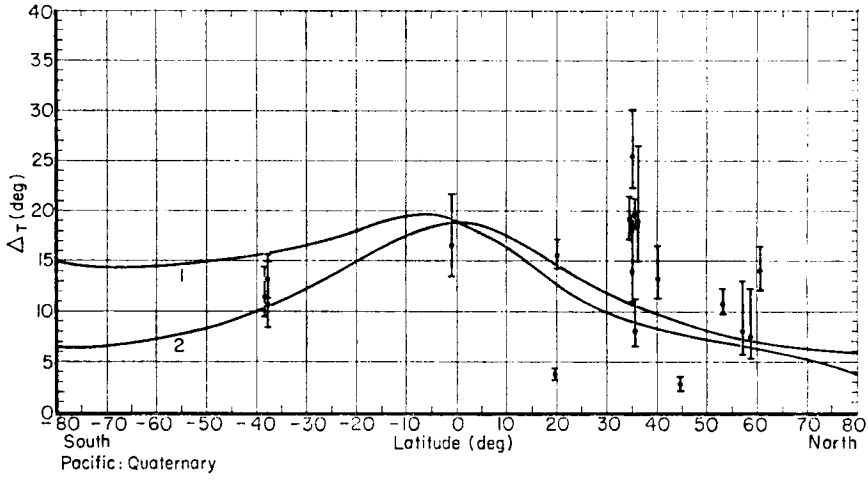


FIG. 2. Quaternary palaeosecular variation data for the Pacific region. Error bars calculated according to Cox (1969a). Data from Table 1, Curves 1 and 2 from Creer (1962).

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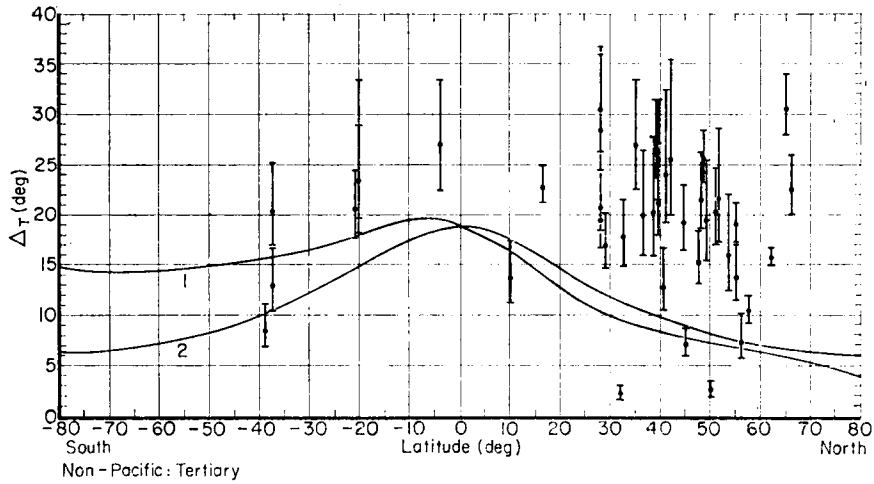


FIG. 3. Tertiary palaeosecular variation data from outside the Pacific region. Error bars calculated according to Cox (1969a). Data from Table 2, Curves 1 and 2 from Creer (1962).

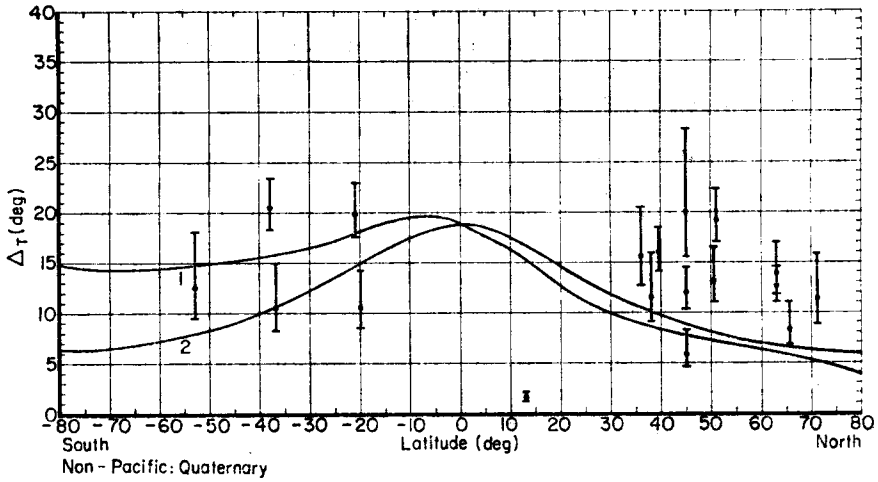


FIG. 4. Quaternary palaeosecular variation data from outside the Pacific region. Error bars calculated according to Cox (1969a). Data from Table 2, Curves 1 and 2 from Creer (1962).

poorly known, as indicated by such ages as 'T–Q' in the tables. Generally, such designated ages are assumed to be young, and for convenience have been included in the 'Quaternary' unless there was some particular reason to include the data with the 'Tertiary' sites.

We have also included in Figs 1–4 two curves based on Creer's (1962a) field rotation model. Curve 1 is the dispersion of the 1945·0 field and curve 2 is the dispersion of the 1945·0 best-fit dipole field.

4. Discussion

Creer & Sanver (1970) have made plots of Δ_β vs. latitude for Cenozoic data, however, they have grouped them according to normally magnetized, reversely magnetized or mixed polarity data with no geographical or temporal groupings.

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They used slightly more rigid selection criteria in choosing their data and thus narrowed the literature down to 26 papers while the data in Tables 1 and 2 represent more than three times that. The principal difference in selection criteria is that Creer & Sanver required sufficient data to be available to complete a 2-tier analysis of variance of either the Watson & Irving (1957) types, or a simpler method described by them. Thus, their diagrams show Δ_β, corresponding to k_β, instead of Δ_T as used in Figs 1–4. For the purposes of this study it was felt that such a condition might be overly restrictive, and that the increase in uncertainty would not outweigh the advantage to be gained by having more data points (where available the results of 2-tier analyses have been included in Tables 1 and 2).

Creer & Sanver (1970) conclude from their study that on a world average throughout the Cenozoic Era, PSV has had about the same latitude dependence as is obtained by rotating the present (1945·0) field about the geographic axis (Creer 1962a, b), except that the north–south asymmetry is averaged out. They find no significant difference between the reversed, normal or mixed polarity sets of data. They make no statements concerning longitudinal distribution of PSV.

Figs 1–4 show considerable scatter of points, making their interpretation difficult. Without, for the moment, making allowances for the statistical as well as other uncertainties in the data, we shall discuss the implications of the trends displayed in these figures.

First of all, the data on all four figures show no clear systematic dependence on latitude. Some, but by no means all, of the data are consistent with one or the other of the curves of Creer (1962a) included in the diagram.

On comparing the Pacific region data with the non-Pacific region data (i.e. comparing Figs 1 and 2 with 3 and 4), it would appear that there has been only a very slight tendency for PSV to be lower in the Pacific region than elsewhere during the Cenozoic Era. This contrast between Pacific and non-Pacific PSV is clearer in the Southern Hemisphere, where there are fewer points on all four diagrams displaying less scatter amongst themselves, than in the Northern hemisphere.

It is also possible to discern a somewhat greater tendency for a smaller Pacific area PSV during the Tertiary than during the Quaternary, though given the probable errors involved, this difference may not be significant.

For the purposes of this analysis, one should perhaps redefine the boundaries of the Pacific low region so as to exclude some of the surrounding continental margins. From Table 1 and Figs 1 and 2 it is clear that the highest PSV values for the Quaternary in the Pacific region are for sites in Japan, and for the Tertiary period for sites in western North America as well as Japan. PSV data for coastal Alaska and the Aleutian Islands for both periods appears to be lower than for the non-Pacific hemisphere at the same latitude while Alaskan sites inland from the Pacific rim have PSV values comparable to the non-Pacific region.

If the changes suggested above are made in Figs 1–4 the low PSV in the Pacific region becomes much more pronounced than previously; the north–south difference in Pacific–non-Pacific PSV contrast is also reduced. However, so few data points are left in the Pacific region after doing this that the significance of the conclusion becomes questionable.

In order to justify the above rather subjective discussion, two kinds of statistical arguments can be applied to draw quantitative conclusions about the reality of the Pacific PSV low.

In order to eliminate the problem of trying to compare two sets of data, namely Pacific and non-Pacific area data, which also have within each set a possible latitude dependence, a 'paired data' comparative analysis has been carried out (e.g. Freund 1967). By this technique the latitude is divided into a number of segments and the difference, x_i, between the Pacific and non-Pacific region mean angular standard deviation for the ith segment is calculated; that is, $x_i = \bar{\Delta}_{Ni} - \bar{\Delta}_{Pi}$. Then the null

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Table 1

Palaeomagnetic statistical data from the Pacific region selected according to criteria described in the text

Explanation: λ and ϕ are latitude and east longitude of sampling site in degrees. $+\lambda$ and $-\lambda$ indicates North and South latitude respectively. Where values are calculated for widely spaced units an approximate mean location is given. Pole No. refers to palaeopole reference numbers in the data compilation of Irving (1964) (decimal format; e.g. 12–42) or in the compilations of McElhinny (1968a, b, 1969, 1970) (slash format; e.g. 10/24) where available. Age is given in million years before present when known or using standard letter symbols for geologic periods as used in the published compilations referred to above and elsewhere in the literature. N is the number of lava flows or units which were used to calculate the statistics. k_T refers to the Fisher (1953) parameter giving unit treatment to N flows or units. k_ω and k_β are the within-flow and between-flow Fisher parameters calculated according to the method of Watson & Irving (1957). δ and Δ_T are angular standard deviations in degrees (see text) $+\varepsilon$ and $-\varepsilon$ are 95 per cent confidence limits on Δ_T in degrees calculated according to Cox (1969a). α_{95} is the radius in degrees of the circle of 95 per cent confidence about the mean palaeofield direction (Fisher 1953). D and I are the mean declination in degrees east of true north and degrees of dip below or above horizontal (indicated positive or negative respectively) of the palaeofield direction after correction for local tectonic movement. In a few cases some of the parameters have been calculated from published data where complete data were not given.

				Pole			Fisher parameter			Angular (s.d.)						
Section description	λ	ϕ	Reference	No.	Age	N	k_T	k_ω	k_β	δ	Δ_T	$+\varepsilon$	$-\varepsilon$	α_{95}	D	I
Alaska:																
Wrangell mean	62·4	216·7	Bingham 1971		6?	25	26·4			15·5	15·8	+3·8	−2·5	5·7	355·3	83·9
Nunivak Is.	60·1	193·5	Cox & Doell 1964		<1	43	33·5[1]				14·0	+2·4	−1·8			
Mt Griggs	58·3	204·7	Stone 1970		T–Q	5	118·8			6·7	7·4	+4·8	−2·1	7·0	334·8	76·2
Aleutian mean	53	188	Bingham & Stone 1971		T–Q	90	56·0			10·8	10·8	+1·2	−1·0	2·0	359·4	70·4
Mt Edgecumbe	57	224·3	Stone 1970		Q	5	102·5			7·2	7·9	+5·1	−2·2	13·8	334·6	77·5
Shemya intrusive & extr rx	52·7	174·1	Cameron & Stone 1970		10·8–18	7	146·8			6·2	6·7	+3·5	−1·7	5·0	11·3	78·8
British Columbia; Caribou Reg.	51·7	233	Symons 1969	11/17	Tmu	48	44·3				12·2	+2·0	−1·5	3·0	356·4	72·3
Columbia River basalts	46·5	239·5	Campbell & Runcorn 1956	11–106	Tm	11	29				15·0	+5·9	−3·3	9	7	65
Columbia Plateau basalts	44·5	240·4	Watkins 1965a	8/28	14·5–21·3	433	12·8				22·6	+1·2	−1·0	6·4	2·9	62·7
Oregon: Marys Peak Sill	44·5	236·4	Clark 1969	11/24	29·6	26	13·6			21·6	21·9	+5·1	−3·5	8·0	21·1	39·8
Suttle Lake	44·4	238·3	Stone, unpublished		T–Q	20	907			2·6	2·7	+0·7	−0·5	6·3	178·1	−60·9
Abert Rim	42·6	239·8	Watkins 1965b	8/26	~15	16	48				11·7	+3·7	−2·2	5	180	−52
Steens Mt	42·6	241·3	Watkins 1969		15	62	8·3			28·2	28·1	+4·0	−3·1	6·7	331·3	77·1
'Western U.S.'	40		Cox & Doell 1964		<1	26[1]	37·2[1]				13·3	+3·1	−2·1			
Japan: Quat ig rx	36	138	Kumagai *et al.* 1950	12–39	Q	11	18				19·1	+7·5	−4·2	11	359	47
baked clays	35·7	139·8	Watanabe 1958, 1959	12–41	historic[4]	174	17				19·6	+1·6	−1·4	3	356	53
historic flows	35·7	139·8	Kato & Nagata 1953	12–40	historic[5]	12	100				8·1	+3·0	−1·7	4	357	46
N. Izu & Hakone rx	35	139	Nagata *et al.* 1957a, b	12–38	Qp–r	42	10			17·2	25·6	+4·5	−3·3	7	343	51
Volcanics	35	134	Sasajima *et al.* 1968	10/40	Te–o	10	40				12·8	+5·3	−3·4	7·1	33	53
Usami Volcano	35	139	Kono 1968	10/12	Qp	11	33				14·0	+5·5	−3·1	8	338	52
Yamaguchi basalt	34·5	131·5	Domen 1960	12–37	Qp	85	18				19·1	+2·3	−1·8	4	0	57
Kyushu basalts, normal	33	130	Creer & Ispir 1970		Tp	12	31·2	45·6[2]	33·5[2]		14·5	+5·4	−3·1		3	3
Kyushu basalts, reversed	33	130	Nomura		Tp	14	17·3	12·4[2]	19·2[2]		19·5	+6·6	−3·9		3	3
Hawaii: five different islands	21·4	202	Tarling 1965	8/5	~1·2–5·7	38	18		25		19·1	+3·6	−2·6	5	357	30
five series on Hawaii	20	204	Doell & Cox 1965	8/4	0·8	112	27	198	29		15·6	+1·6	−1·4	2·6	5·7	31·1
Kau series	19·5	204·5	Doell 1969	11/1	<0·01	54	479	304	610		3·7	+0·6	−0·4	0·9	354·8	24·1
Galapagos Islands	− 2	268	Cox & Doell 1964		<1	21[1]	24·2				16·5[1]	+5·2	−2·9			
Samoa	−14·7	188·8	Tarling 1966b	8/12	2	16	20		25		18·1	+5·7	−3·5	8	5	−3
Australia: Q'land lavas & dyks	−27·0	152·2	Robertson 1966	8/29	Tu	12	25				16·2	+6·0	−3·4	9	13	−49
Nandewar volcanoes	−30·3	152·2	Wellman *et al.* 1969	11/20	17·5±0·3	34	43·1			12·1	12·3	+2·4	−1·7	3·7	192·4	53·0
Liverpool volcanoes	−31·7	150·2	Wellman *et al.* 1969	11/22	33·4±0·7	36	45·5			11·8	12·0	+2·3	−1·7	3·5	200·4	59·2
Barrington volcanoes	−32·0	151·4	Wellman *et al.* 1969	11/27	51·6±0·7	33	48·5			11·4	11·6	+2·3	−1·7	3·6	193·0	65·5
Victoria, old volcanoes	−38·0	145·5	Irving & Green 1957a, b	11–093	Te–o	15			35	13·7	13·7[6]	+4·5	−2·7	6·8	17·0	−72·9
Victoria, new volcanoes	−38·0	143·5	Green & Irving 1958	12·45	Tp–Qr	32			37	13·2	13·3[6]	+2·7	−1·9	4·8	3·4	−59·8
Victoria	−38	143·5	McDougall *et al.* 1966	8/9	0·5–4·5	11	56				10·8	+4·2	−2·4	6	350	−56
New Zealand: N. Island	−38·5	176	Cox 1969a	11–9	<0·68	22	49	322	49·9		11·6	+3·0	−2·0	4·5	2·0	−63·1
Akaroa Volcano	−43·9	173·0	Evans 1970		8·4–9·1	70	13·8	10·4[2]	19·4[2]	21·8	21·8	+2·9	−2·3	4·7	177·3	54·9

[1] N read from published diagrams. Δ_T calculated from values of Δ for VGP's given in reference paper according to Cox (1962). k_T calculated from Δ_T by equation (3). No mean direction available.

[2] k_ω and k_β calculated from δ_ω and δ_β using equation (3) where δ_ω and δ_β are related according to: $\delta_\beta^2 = \delta^2 - \frac{\delta_\omega^2}{N}$

[3] Mean vector directions not available.

[4] Dated at 4000 B.C. to 1600 A.D.

[5] Dated to 325 to 1779 A.D.

[6] Δ_T calculated from k_β since k_T not available.

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Table 2

World-wide palaeomagnetic statistical data from outside the Pacific region selected according to criteria described in the text

See caption to Table 1 for explanation of symbols

Section description	λ	ϕ	Reference	Pole No.	Age	N	Fisher parameter k_T	k_ω	k_β	ϑ	Angular (s.d.) Δ_T	$+\varepsilon$	$-\varepsilon$	α_{95}	D	I
Norway: Jan Mayen Lavas	71·1	351·8	Fitch *et al.* 1965	8/3	Qr	10	52				11·2	+4·6	−2·5	7	340	83
Iceland: volcanics	66	336·5	Kristjansson 1968	10/39	Te–m	60	12·9			22·5	22·6	+3·3	−2·5	5·3	0·5	73·0
Jökuldalur	65·3	344·8	Wensink 1964		Tp–Qp	16	94·0			8·1	8·3	+2·6	−1·6	3·8	10·8	79·0
Miocene lavas	65·2	340	Hospers 1953–54, 1955	11–108	Tm	102	7			10·7	30·6	+3·3	−2·7	5·5	1·5	77·8
Sweden: post-glac vrvs I	63·1	17·7	Bancroft 1951	12–12	historic[1]	46	42				12·5	+2·1	−1·6	3·3	357·4	73·4
post-glac vrvs II	63·1	17·7	Griffiths 1953, 1955	12–13	historic[2]	29	34				13·9	+3·0	−2·1	4	2	75
Faeroe Islands: lavas	61·8	353·0	Tarling 1968, 1970		53–59	253	30[3]				15·8	+1·0	−0·9			
Scotland: Skye lavas & intr	57·4	353·7	Khan 1960	11–012	Te–o	53	60				10·4	+1·6	−1·2	3	186	−60
N. Ireland: Antrim rx	55·1	353·6	Wilson 1959	11–024	Te–o	89	18				19·1	+2·2	−1·8	4	188	−63
Ireland: Garren Pt.	55·1	353·9	Wilson 1970		T	25	26·4				13·8	+3·3	−2·2	5·1	184·7	−54·3
British Isles: Tert ig rx	56	355	Irving 1964	11–026	Te–o	11	123				7·3	+2·9	−1·6	4	6	63
Tert dykes	53·6	357·1	Dagley 1969	11/28	Te	11	25·9			15·2	15·9	+6·2	−3·5	9·1	169·5	−61·4
Poland: Low Silesia[4]	51·1	15·5	Birkenmajer & Nairn 1969	11/16	To–Qp	56	18·0			19·0	19·1	+2·9	−2·2	4·6	184·3	−65·6
Wzar andesite dykes	49·2	20·2	Birkenmajer & Nairn 1969	11/21	Tm	15	17·5				19·4	+6·1	−3·8	9·4	191·5	−73·2
Czechoslovakia: T & Q ig rx	48·5	19	Nairn 1966	10/24	Tm–p	94	10				25·6	+2·9	−2·3	5	11	63
volcanics	48	21	Nairn 1967	9/24	Tm	33	14				21·7	+4·4	−3·1	7	359	64
Hungary	47·5	19·3	Dagley & Ade-Hall 1970		T	33	28			15	15·3	+3·1	−2·2	4·8	2·2	61·0
Germany: Gottingen volcanics	51·4	9·8	Schult 1963; Irving 1964	11–115	Tm–p	15	14				21·6	+7·0	−4·3	11	184	−64
Lausitz	51	14·7	Nairn & Vollstadt 1967	10/35	To–m	27	16				20·3	+4·6	−3·2	7	200	−63
Reinl. Pf. ig rx	50·6	7·5	Nairn 1962	11–113	To–Q	22	38			9·5	13·1	+3·4	−2·2	5	28	62
Suevites, N. Ries	49·9	10·5	Peterson *et al.* 1965	8/30	14·8±7·0	12	997				2·6	+1·0	−0·6	2	191	−60
France: Ch. des Puys, Louch	45	3	Bonhommet & Zähringer 1969	11/4	Qr	10	193				5·8	+2·4	−1·3		114	57
Ch. des Puys, Bruhnes	45	3	Doell 1970a		<0·7	31	47·0	199·3	47·8		11·9	+2·5	−2·4	3·8	357·8	60·3
Ch. des Puys, Matuyama	45	3	Doell 1970a		>0·7	10	16·9	144·6	17·3		19·9	+8·3	−4·5	12·1	357·9	49·4
Plateau du Velay	45	3·8	Bobier 1969	11/12	<2·5	28	130		49		7·1	+1·6	−1·1	4	9·5	58
Coiron lavas	44·6	4·6	Wensink 1970		Tm–Qp	36	17·6				19·2	+3·7	−2·7	5·8	9·5	54·9
Sardinia: Logudoro lavas	40·5	8·5	Bobier & Coulon 1970		Tm	21	40				12·8	+4·0	−2·2	10	328·5	43
Italy: Mr. Etna Lava	37·7	15·0	Chevallier 1925	12–09	historic[5]	11	50				11·5	+4·5	−2·5	7	356	69
Portugal: Lisbon volcanics	38·8	350·8	Watkins & Richardson 1969	10/42	Te	12	15·9	63·3	17·1		20·3	+7·5	−4·3	11·2	346·7	37·2
Madiera volcanics	32·5	347	Watkins *et al.* 1966b	8/19	⩽Tm	29	20·9				17·7	+3·8	−2·7	6·0	3·5	47·5

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USSR:																
Georgia, volc & seds	42	43	Vekua 1961	11–117	Te	11	10				25·6	+10·0	−5·6	15	356	45
Azerbardzhan	41	49	Khramov & Andreyeva 1964	8/13	Tp	13	11·4				24·0	+8·4	−4·9	13	12	49
Turkmenian seds	39·5	55	Khramov 1958	11–044	Tpa–o	26	10				25·6	+6·0	−4·0	9	34	44
Baku Beds	39·5	53·2	Khramov 1958	11–044	Q	47	10				25·6	+6·0	−4·0	9	11	54
Apsh. & Akch. stages	39	53	Khramov 1958	12–30	Tp–Qp	47	9				27·0	+4·5	−3·3	8	359	42
Algeria:																
Massif de Cavallo	32	5	Bobier & Robin 1969	11/19	Tm	13	1381				2·2	+0·8	−0·5	2	1	54
Aden:																
volcanics	12·8	45·0	Irving & Tarling 1961	12–42	T–Q	12	2740	52	340		1·5	+0·6	−0·3	2·7	353·1	24·0
Ethiopia:																
flood basalts	9·5	38·7	Brock *et al.* 1970		Te–m	20	34·9	25·8	86·4		13·7	+3·8	−2·4	5·6	7·2	7·2
Canary Islands:																
Lanzarote	29	346·5	Watkins *et al.* 1966a	8/18	Tm–Qr?	38	22·7				17·0	+3·2	−2·3	5·0	10·9	34·6
Gran Canaria	28	344	Watkins *et al.* 1966a	8/17	Tm–Qr?	39	7·1				30·4	+5·6	−4·1	9·3	10·1	37·6
Teneriffee	28	343	Watkins *et al.* 1966b	8/16	Tm–Qr?	46	14·9				21·0	+3·5	−2·6	5·6	3·4	39·8
Gomera	28	342·5	Watkins *et al.* 1966a	8/15	Tm–Qr?	18	8·1				28·5	+8·3	−5·2	13·0	7·9	33·1
Hierro	28	342	Watkins *et al.* 1966a	8/14	Tm–Qr?	33	17·3				19·5	+3·9	−2·8	6·2	6·9	45·6
Cape Verde Islands	16·5	336	Watkins *et al*, 1968	10/32	Tm	138	12·6				22·8	+2·1	−1·7	3·5	1·6	22·2
Fernando Noronha	− 3·9	327·6	Richardson & Watkins 1967		Tm	24	9·1				26·9	+6·5	−4·4	10·4	0·8	−11·9
Réunion Is:																
Gp. I lavas	−21	55·5	Chamalaun 1968	10/6	0·6–1·0	49	16·7				19·8	+3·2	−2·4	5·1	6·3	−39·2
Gp. III lavas	−21	55·5	Chamalaun 1968	10/8	1·9–2·0	38	15·4				20·6	+3·8	−2·8	6·1	181·1	35·4
Mauritius:																
late lavas	−20·3	57·5	McDougall & Chamalaun 1969	11/10	0·17–0·70	13	58·4			10·1	10·6	+3·7	−2·2	5·4	359·3	−45·7
early lavas	−20·3	57·5	McDougall & Chamalaun 1969	11/11	1·96–3·44	10	11·8			22·5	23·6	+9·8	−5·4	14·7	4·9	−45·2
older volc series	−20·3	57·5	McDougall & Chamalaun 1969	11/13	5·26–7·88	26	11·9			23·1	23·5	+5·5	−3·7	8·5	358·3	−42·2
Tristan de Cunha & Inaccess.	−37	347	Creer 1964	8/2	Qr	10	10	58			10·6	+4·4	−2·4	7·2	0	−50
Heard Is.	−53	73·5	Irving *et al.* 1965	8/10	T–Qr	9	42	113	53	11·2	12·5	+5·6	−2·9	8	35·3	−62
Nevada:																
Lousetown Fm	39·4	240	Heinrichs 1967	9/11	1·1–1·9	23	14·1				21·6	+5·4	−3·6	8·4	22·7	45·3
New Mexico:																
Rio Grand	36·5	254·3	Ozima *et al.* 1967		3·7–4·4	14	16·6			19·2	19·9	+6·7	−4·0	10·0	351·2	47·4
Valles Caldera	35·9	253·4	Doell *et al.* 1969	11/8	0·43–1·37	16	26·8			15·2	15·6	+4·9	−3·0	7·3	2·9	53·3
N. Mex. & Arizona:																
lavas & seds	35·2	248·4	Kono *et al.* 1967	9/13	3·7–4·5	24	9				27·0	+6·6	−4·4	10	177	−51
Argentina:																
Neuq.-Mend. basalt	−37·5	290	Valencio & Creer 1968	10/21	0·45–27·6	24	16				20·3	+4·9	−3·3	8	352	−60
Neuquen lavas	−38	290	Creer 1958	12–48	Q	58	15·4				20·6	+3·0	−2·3	5·0	1	−61
Neuquen	−39	294·0	Valencio 1965a, b		Tm–p	16	91·1			8·2	8·5	+2·7	−1·6	3·9	337·0	−59·6
basalts[6]	−37·5	290	Creer & Valencio 1970	11/15	To–Q	18	38·8	16·4	54·2		13·0	+3·8	−2·4	6	359	−59

[1]Dated by carbon-14 to the period 0–1000 A.D.

[2]Dated by carbon-14 to the period 1100 B.C. to 750 A.D.

[3]Δ_T meaned from five values for different series read from diagram. *D* and *I* not available.

[4]Statistics calculated from site mean statistics given in paper; several sites omitted because of large α_{95}.

[5]Historically dated from 394 B.C. to 1911 A.D.

[6]These data include some lavas from the above three entries.

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Table 3

'Paired data' statistical analysis of palaeosecular variation data

	Q	Q′	T	T′	TQ	TQ′
$\bar{x}$	−1·24	−1·63	+4·16	+5·16	+1·21	+1·55
s	5·94	5·70	3·12	4·19	4·67	5·37
n	6	5	5	4	7	7
t	−0·512	−0·642	2·98	2·463	0·683	0·765
$t_{0\cdot05,\,n-1}$	2·015	2·132	2·132	2·353	1·943	1·943
H_0	accept	accept	reject	reject	accept	accept

Explanation: $\bar{x}$, s, n, t, H_0 are explained in text. $t_{0\cdot05,\,n-1}$ is the t distribution as found in statistical tables. H_0 accepted or rejected according as $t <$ or $> t_{0\cdot05,\,n-1}$. Q, T and TQ refer to Quaternary Tertiary, and combined Tertiary–Quaternary PSV data from Tables 1 and 2. Unprimed columns indicate data grouped as in Figs 1–4; primed (′) columns indicate geographical grouping altered as described in text to reduce the area included within the Pacific region.

Table 4

Statistical analysis of palaeosecular variation data ignoring latitudinal dependence and using normal distribution

	Q	Q′	T	T′	TQ	TQ′
$\bar{\Delta}_P$	12·95	11·16	16·08	14·73	14·56	12·95
$\bar{\Delta}_N$	13·02	14·03	19·09	18·82	17·27	17·09
z_P	5·86	3·97	5·12	4·43	5·64	4·46
z_N	5·13	5·75	6·96	6·74	6·99	6·83
n_P	18	10	19	10	37	20
n_N	18	26	42	51	60	77
z	0·038	1·701	1·892	2·42	2·09	2·59
H_0	accept	reject	reject	reject	reject	reject

Explanation: All quantities are explained in text or in Table 3. H_0 accepted or rejected according as z is $<$ or $> 1\cdot67 = z_{0\cdot05}$ for one tailed test.

hypothesis, H_0, that there is no difference in PSV between the two regions, is tested against the alternate hypothesis, H_1, that PSV is greater outside the Pacific region. Since the number of paired data available is small, it is necessary to use a one-tailed t-test by calculating the statistic

$$t = \frac{\bar{x}}{s}\sqrt{n},$$

where $\bar{x}$ is the mean of n values of x_i having standard deviation s.

For this analysis, the geographic latitude of the data shown in Figs 1–4 was divided up into 20° segments with the central boundary at the equator. Smaller intervals were not feasible because the paucity of available data would have left many data unpaired. Intervals wide enough to leave no data unpaired would make n unreasonably small.

The calculations (Table 3) were carried out using six different sets of data; Quaternary, Tertiary, and Tertiary–Quaternary combined, for both original (as presented in Figs 1–4) and a modified geographic distribution as described above. Only for the Tertiary data is there a clear-cut case at the 5 per cent level for rejecting H_0 and concluding that the Pacific low is real; this holds even without decreasing the area originally included in the Pacific region. In all the other cases it appears that there is no significant difference in PSV between the two regions.

Since using the paired data technique necessitated ignoring much data which could not be paired, and since the data show no obvious latitude dependence, a second statistical analysis was carried out assuming each of the data in Figs 1–4

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represent random samples from a population possessing a normal distribution; that is, it is presumed that there is no latitude dependence of Δ. Thus, the statistic

$$z = \frac{\bar{\Delta}_N - \bar{\Delta}_P}{\sqrt{[(s_N^2/n_N) + (s_P^2/n_P)]}}$$

is calculated, where $\bar{\Delta}$ is the mean of n values of Δ having standard deviation s, and subscripts $_N$ and $_P$ refer to data from the non-Pacific and Pacific regions. The null hypothesis H_0 is then tested at the 5 per cent level as before by comparing calculated z values with the normal distribution (Table 4).

This method, applied to the six cases enumerated above, gives a much more definite indication of the existence of the Pacific PSV low. Only in the case of the geographically unmodified Quaternary data are we forced to accept the null hypothesis, H_0. In all other cases there appears to be sufficient reason to conclude that PSV is less in the Pacific region than elsewhere.

5. Conclusions

Until more suitable definitive palaeomagnetic data become available, any statements made about the Pacific region PSV low must remain provisional. Based on the above subjective discussions and statistical analyses, we may draw the following very tentative conclusions.

(1) During the Cenozoic Era secular variation of the geomagnetic field has been somewhat less in the region containing the Pacific Ocean than elsewhere.

(2) The contrast in PSV between the Pacific and non-Pacific regions of the Earth is enhanced if the west coast of North America, mainland Alaska, and Japan are excluded from the Pacific region.

(3) This geographical contrast in PSV was slightly greater during Tertiary than during Quaternary times.

It is clearly very important to obtain more information on the behaviour of the Pacific secular variation low back through time, because of the implication with respect to the asymmetry of the deep structure of the Earth.

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